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Water vapour evaporates from mesophyll cells to the intercellular spaces near stomata as a result of active transpiration. The water vapours are then transpired through the stomatal pores. Loss of water from mesophyll cells causes a decrease in water potential. So, water moves as a pull from cell to cell along the water potential gradient. This tension, generated at the top (leaf) of the unbroken water column, is transmitted downwards from petiole, stem and finally reaches the roots. The cohesion theory is the most accepted among the plant physiologists today.

11.6 Transpiration

Water absorbed by roots ultimately reaches the leaf and gets released into the atmosphere in the form of vapour. Only a small fraction of water (less than 5%) is utilized in plant development and metabolic process.

The loss of excess of water in the form of vapour from various aerial parts of the plant is called **transpiration**. Transpiration is a kind of evaporation but differs by the involvement of biological system. The amount of water transpired is astounding (Table 11.4). The water may move through the xylem at a rate as fast as 75cm /min.

Table: 11.4 Rate of Transpiration in some plants

Plant	Transpiration per day
Corn plant	2 Litres
Sunflower	5 Litres
Maple tree	200 Litres
Date palm	450 Litres

Activity

Select a leafy twig of fully grown plant in your school campus. Cover the twig with a transparent polythene bag and tie the mouth of the bag at the base of the twig. Observe the changes after two hours and discuss with your teacher

11.6.1 Types of Transpiration

Transpiration is of following three types:

1. Stomatal transpiration

Stomata are microscopic structures present in high number on the lower epidermis of leaves. This is the most dominant form of transpiration and being responsible for most of the water loss (90 - 95%) in plants.

2. Lenticular transpiration

In stems of woody plants and trees, the epidermis is replaced by periderm because of secondary growth. In order to provide gaseous exchange between the living cells and outer atmosphere, some pores which looks like lens-shaped raised spots are present on the surface of the stem called **Lenticels**. The loss of water from lenticels is very insignificant as it amounts to only 0.1% of the total.

3. Cuticular transpiration

The cuticle is a waxy or resinous layer of **cutin**, a fatty substance covering the epidermis of leaves and other plant parts. Loss of water through cuticle is relatively small and it is only about 5 to 10 % of the total transpiration. The thickness of cuticle increases in xerophytes and transpiration is very much reduced or totally absent.